ORION AI INNOVATION PROJECT AWARDS 2026 · IIT MADRAS RESEARCH PARK

TerraShield AI

One digital view of the land.

An AI-powered 3D land intelligence platform for disaster risk and recovery — so decisions before, during and after disasters are faster, clearer and evidence-linked.

TEAM

DUWO · Abhinav S and Goutham K

Chennai Institute of Technology · BE CSE · 1st Year

OPENING SCENE

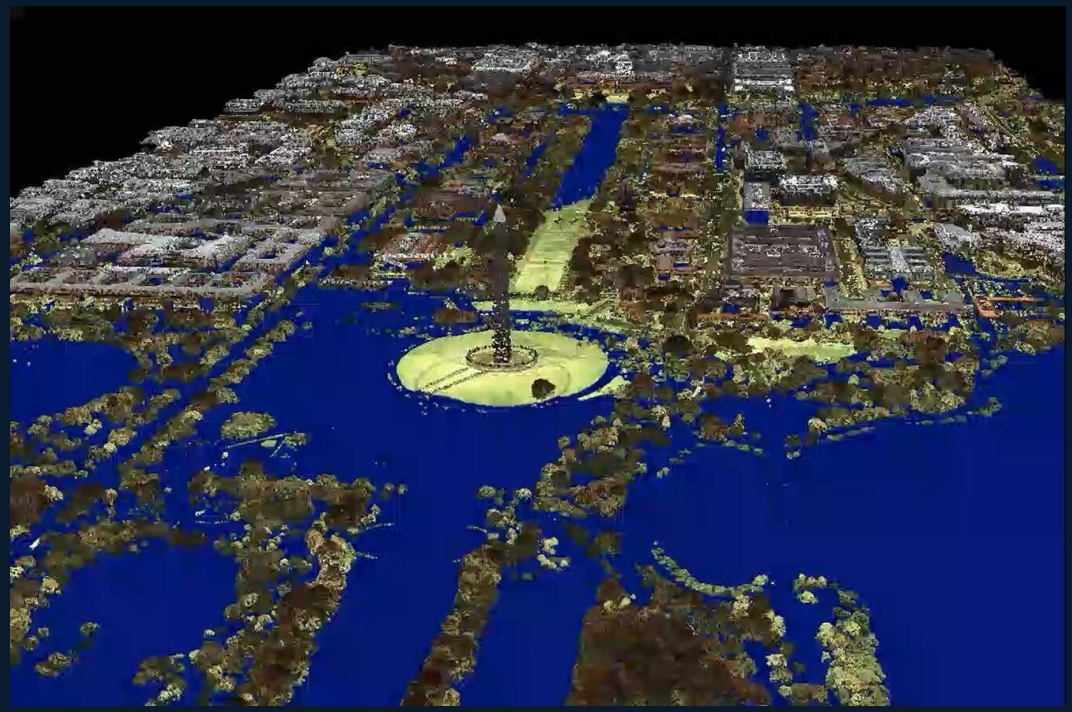
11:47 PM. Your phone buzzes.

A cloudburst is forecast over your city. Landfall: 18 hours.

FORECAST ISSUED
Which ward floods first?

WATER RISES
Which road do you close first?

TEAMS DEPLOY
Where do you send them first?



3D flood simulation · USGS — concept reference

You have 9 data portals · 4 PDF reports · 3 departments — **and one night.**

THE PROBLEM

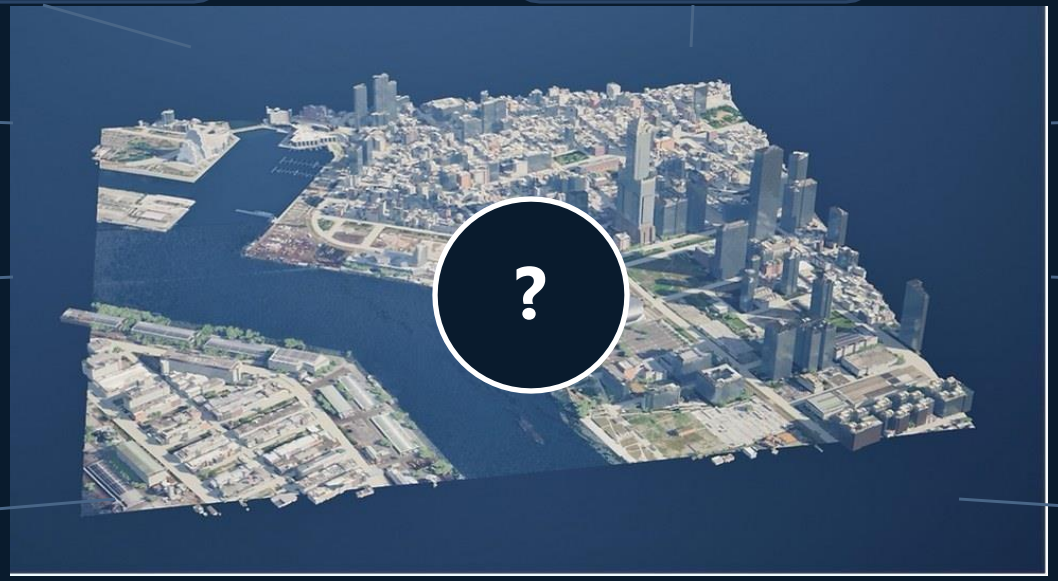
One city. A dozen datasets. Zero shared views.

Land-use rules

Hazard records

Terrain

Infrastructure



Soil & water bodies

Elevation

Drainage

Reports & portals

How do all these layers relate — HERE?

The hard part isn't finding a layer — it's relating them at one location. **Result: slower, opaque decisions when hours matter most.**

THE COST OF FRAGMENTATION

The decision tax

9+

portals, logins & PDFs to check
before a single go / no-go decision

4–6 hrs

to assemble a preliminary view
of just one site or ward

0

unified, evidence-linked views
of one place — anywhere

Try it on the room: "How many websites would YOU need to open before approving a building on an empty plot?"

Most people pause. That pause is the problem we're solving.

THE SOLUTION

TerraShield AI — one digital view of the land



IN ONE SCREEN

▦ Every layer on one location — terrain, water, roads, hazards

⚑ Click any flag → the why, the source, the data limits

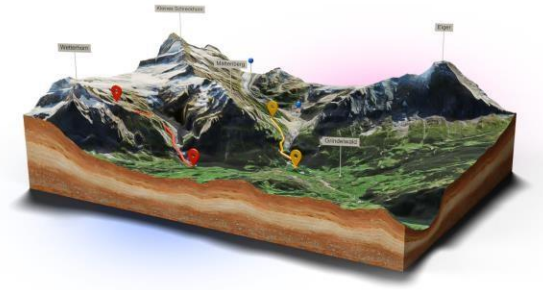
𝔈 Rotate, zoom, switch between suitability · flood · infrastructure · recovery

⚠ Uncertainty shown honestly — never hidden

12 DATASETS → 1 VIEW **HOURS → MINUTES**

HOW IT WORKS

From data to action in five moves



THE 3D VIEW IS THE ANCHOR

1

Select
a plot or area

2

Assemble
available geo & hazard data

3

Analyse
land traits & mapped risks

4

Explore
findings in interactive 3D

5

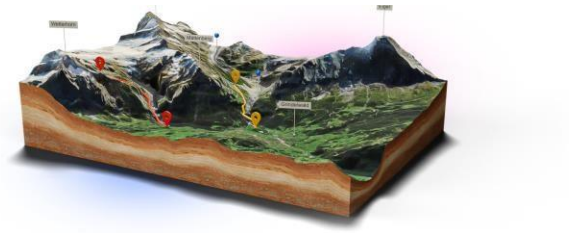
Report
evidence-linked assessment

Every step stays on the same location profile — nothing gets lost between assessment, preparedness and recovery.

ONE PLATFORM

Three intelligence modes. One location profile.

THE SAME PLACE, THREE CRITICAL MOMENTS



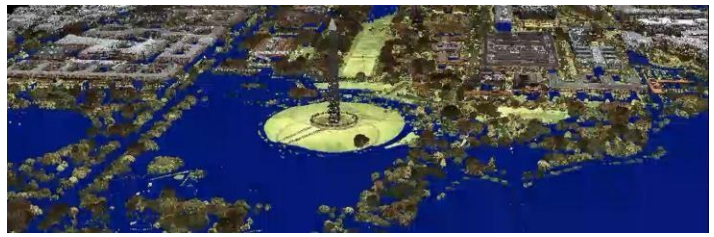
BEFORE DEVELOPMENT

- ✓ Preliminary land-suitability indicators
- ✓ Mapped environmental constraints
- ✓ Terrain context + verification checklist



DISASTER PREPAREDNESS

- ✓ Flood & waterlogging information
- ✓ Vulnerable infrastructure mapped
- ✓ Areas requiring attention flagged



AFTER A DISASTER

- ✓ Organized, verified damage reports
- ✓ Affected infrastructure overview
- ✓ Inspection & repair priorities

Prioritization depends on data + validation.

UNDER THE HOOD

Responsible AI wrapped around a geospatial core



⚠ **AI/ML only for specific, validated tasks — never an LLM alone as a disaster predictor**

GUARDRAILS BY DESIGN

GIS is the ground truth
Every AI finding sits on mapped spatial evidence.

Document-reading AI
Extracts from official reports — always with source references.

Uncertainty flags
Missing, outdated or low-confidence data is shown, not hidden.

Human-centred support
Explains findings and suggests follow-up — humans decide.

RESPONSIBLE AI

Trust by design, around a geospatial core

GIS grounded
Risk flags anchored to terrain, layers and spatial relationships — not generated text.

Validated models only
AI/ML for specific tasks when labelled data and validated models exist.

✓
GEOSPATIAL CORE

Source-linked documents
Official reports read by AI with citations — every claim traceable.

Uncertainty & oversight
Weak data is flagged; support stays transparent and human-led.

A language model alone is never presented as a flood or disaster predictor.

COMPETITION

Integration is the differentiator

CAPABILITY	TerraShield AI	Typical single-purpose tools
Land-to-recovery continuity one profile across assessment, preparedness & recovery	✓	—
Explainable findings every flag linked to its 'why' and its sources	✓	—
Uncertainty awareness missing / outdated / low-confidence data is flagged	✓	—
Accessible to non-experts 3D exploration for citizens & field teams	✓	—
Deep single-domain models e.g. specialized flood forecasting at national scale	—	✓

Honest positioning: our edge comes from integration & presentation — we don't claim no comparable systems exist.

WHO IT'S FOR

Built for people who make place-based decisions



Urban & regional planners

Compare terrain, land-use constraints and infrastructure context during preliminary planning.



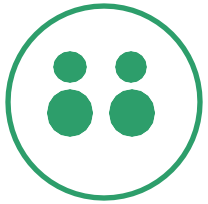
Builders & infrastructure planners

Review mapped constraints and identify what needs engineering verification.



Authorities & disaster teams

Explore hazard information, vulnerable infrastructure and organized inspection priorities.



Communities & citizens

Understand location-based information through an accessible visual interface.

Decision support only — never a replacement for engineering surveys, emergency services, statutory approvals or expert inspection.

FEASIBILITY

A feasible prototype, built in layers

MINIMUM VIABLE PROTOTYPE

1. One pilot area + authoritative data sources
2. Interactive 3D geographic interface
3. A small number of relevant data layers
4. Transparent risk flags in 3D
5. One evidence-linked assessment report

PLANNED —PROTOTYPE BUILT . DEVELOPMENT IN PROGRESS

TECHNOLOGY STACK

CesiumJS / Three.js
interactive 3D

QGIS + Python
geospatial analysis

Public datasets
elevation · land · water · hazard

Validation loop
records + expert feedback

Web dashboard
map UI + report generator

STATUS & ROADMAP

Concept today — validation is the next proof point

TODAY

Concept
NOW — student innovation proposal

Pilot & data
select area + authoritative sources

3D interface
interactive location view

Integrate layers
a few relevant datasets

Source-linked reports
preliminary assessments

Validation
known records + expert feedback

HOW WE'LL EVALUATE

- Source traceability
- Data coverage
- Missing-info detection
- Agreement with known hazard records
- Report usability

DEMO PROTOTYPE : https://terrashieldmain.netlify.app/. (DEVELOPMENT IN PROGRESS)

From fragmented data
to decisions you can defend.

TerraShield AI makes place-based evidence easy to explore, question and act on — before, du and after disasters.



ONE VIEW. EVERY LAYER.

Future scope: validated flood simulation · landslide & cyclone extensions · verified field observations · AR visualization

OUR ASK **48 hours · one flood-prone ward · open data.**

Help us prove the first digital twin — we'll bring the working prototype.

TEAM

Abhinav S · TECHNICAL LEAD + PROTOTYPE DEVELOPER · Goutham K · PRODUCT RESEARCH + PRESENTATION LEAD

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— Thank you!